AnemiaFusionNet

A Multimodal Feature Fusion Framework for Region-Aware Anemia Detection

Technical Evaluation Report

Modalities: Ocular Imaging · Clinical CBC Biomarkers · NFHS-5 Geographic Prevalence Prior

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1. Executive Summary

AnemiaFusionNet is a tri-modal deep learning framework designed to detect anemia by combining three complementary information sources: ocular (conjunctival) images, clinical complete blood count (CBC) biomarkers, and a region-level epidemiological prior derived from the National Family Health Survey (NFHS-5). The framework employs independent modality-specific encoders — an EfficientNet-B0 vision backbone, a tabular MLP with categorical embeddings, and a geo-risk MLP — whose 256-dimensional outputs are fused via a multi-head selfattention transformer operating over a 4-token sequence ([CLS], image, clinical, geo).

This report is based on the project's current per-configuration metrics export (accuracy, precision, recall, F1, and AUC computed directly from model predictions — see Figure 6, Section 6), which is corroborated independently by the confusion matrix (Section 7, Figure 2) and the ROC curve (Section 7, Figure 1): all three artifacts agree with one another for the Full Fusion configuration. Read against these figures, clinical CBC biomarkers alone provide a strong predictive signal, a purely image-based classifier collapses to majority-class prediction on the currently available cohort size, and the full tri-modal fusion (image + clinical + geographic prior) achieves the highest ROC-AUC of any configuration (93.98%), ahead of Image + Clinical (91.73%) and Clinical-only (89.85%). Because the tri-modal dataset was constructed via a simulated linkage procedure rather than sourced from a single cohort with all three modalities collected on the same patients, the quantitative results in this report should be interpreted as a technical feasibility demonstration rather than as clinically validated performance.

2. Introduction and Motivation

Anemia remains a widespread public health concern, with particularly high prevalence recorded across Indian states in national health surveys. Conventional diagnosis relies on laboratory CBC testing, which requires venipuncture, reagents, and trained personnel — infrastructure that is not uniformly available in low-resource or rural settings. Non-invasive screening from conjunctival pallor images has been explored as an alternative, but single-modality image classifiers tend to struggle when trained on the small, publicly available ocular-image cohorts, since visual pallor is a comparatively subtle and noisy signal relative to direct blood biomarkers.

AnemiaFusionNet's motivating hypothesis is that combining (a) a direct blood biomarker signal, (b) a non-invasive visual signal, and (c) a regional epidemiological base-rate prior through a learned attention mechanism can produce a more robust and more clinically interpretable classifier than any single modality alone, while retaining a pathway toward non-invasive, image-led screening in settings where CBC access is limited.

3. Dataset and Modalities

Three modalities were sourced from three independent datasets and combined into a unified patient-level record:

Modality	Source	Data Type	Features	Purpose
Image (Ocular)	EYES-DEFY-ANEMIA (Kaggle)	RGB image	Conjunctiva/sclera photos — 95 Indian + 123 Italian patients (218 total)	Fine-tune vision backbone for conjunctival pallor
Clinical (Tabular)	Diagnosed CBC Dataset (Kaggle)	Numeric + categorical	13 CBC markers (RBC, WBC, HCT, MCV, MCH, MCHC, PLT, PDW, PCT, LYMp, LYMn, NEUTp, NEUTn) + Gender + region_type	Capture pathological blood biomarkers
Geographical	NFHS-5 (National Family Health Survey)	Statistical prior	State index + normalized anemia prevalence score (0–1)	Inject region-level epidemiological base rate

The final cohort used for tri-modal training and evaluation is bounded by the smaller image dataset (218 conjunctiva images), which is a material constraint discussed further in Section 10.

4. Data Linkage and Simulation Methodology

Data constraint disclaimer

No publicly available dataset combines eye images, full CBC tabular features, and patient location for the same cohort. A simulated data-linkage pipeline was built to bridge the three sources and demonstrate technical feasibility of the fusion architecture.

4.1 Conceptual Summary

- True target labels: computed from each patient's real hemoglobin (Hb) measurement in the ocular-image dataset, against WHO age/gender-specific thresholds.
- Matching blood tests: a CBC donor record is randomly drawn from the diagnosed-CBC dataset, conditioned to match the patient's anemic/healthy label.
- Assigning locations: an Indian state is randomly sampled proportional to population share, and the corresponding real NFHS-5 state-level prevalence score is attached.

4.2 Technical Implementation

The linkage is deterministic per patient index i (seeded as random_state = idx):

Label engineering: $y_i = \mathbb{1}[\, Hb_i < \theta\_WHO(Age_i, Gender_i) \,]$, using ground-truth hemoglobin, age, and gender from the image dataset.

Clinical linkage: x_cbc,i is drawn uniformly from CBC records sharing label y_i; the continuous HGB column is dropped from the donor record to prevent target leakage, while WBC, RBC, MCV and other indices are retained.

Geographic prior: a state s_i is drawn from a categorical distribution weighted by state population share; the prior $R(s_i)$ is the min–max normalized NFHS-5 prevalence value for that state, scaled to [0, 1].

Because the clinical and geographic values are sampled independently of the true patient (conditioned only on the coarse binary label or on population share), any performance gain attributed to the clinical or geo branches reflects how well those modalities separate the label distribution in their own donor pools — not a validated per-patient biological or geographic relationship. This is an important distinction for interpreting the ablation results in Section 6.

5. System Architecture

AnemiaFusionNet uses hybrid feature-level fusion: three parallel modality encoders each project their input into a shared 256-dimensional embedding space; these embeddings, together with a learnable [CLS] token, form a 4-token sequence that is passed through a multi-head self-attention transformer.

5.1 Pipeline

- Conjunctiva image → Vision Encoder (EfficientNet-B0) → 256-d embedding
- Clinical CBC metrics → Tabular Encoder (MLP + categorical embeddings) → 256-d embedding
- Geographic state + NFHS-5 prior → Geo-Risk Encoder (state embedding + MLP) → 256-d embedding
- [CLS] token + 3 modality embeddings + modality-type segment embeddings → 4-layer, 8-head self-attention (d_model = 256, dim_ff = 512)
- Pooled [CLS] output → 2-layer MLP classification head → Anemic vs. Healthy

5.2 Encoder Specifications

Branch	Backbone	Input	Output
Vision	EfficientNet-B0 (ImageNetpretrained)	Conjunctiva RGB image, classification head replaced	256-d projection
Clinical	MLP + categorical embeddings	13 numeric CBC features + Gender + region_type	256-d projection
Geo-Risk	16-d state embedding + MLP	State index + 1-d normalized NFHS-5 prevalence score	256-d projection
Fusion	4-layer / 8-head Transformer	[CLS, Image, Clinical, Geo] token sequence	Pooled [CLS] → classification MLP

6. Ablation Study Results

Four configurations were compared on the held-out test set to isolate the contribution of each modality. The table below is taken from the project's current per-configuration metrics export (Figure 6, Section 10). This export is internally consistent with the other reported artifacts for the Full Fusion configuration: its accuracy/precision/recall/F1 can be reproduced exactly from the confusion matrix in Section 7 (Figure 2), and its AUC matches the plotted ROC curve's legend (Figure 1).

Configuration	Accuracy	Precision	Recall	F1	ROC-AUC
Image-only (EfficientNet-B0)	42.42%	42.42%	100.00%	59.57%	50.00%
Clinical-only (Tabular MLP)	90.91%	86.67%	92.86%	89.66%	89.85%
Image + Clinical	87.88%	85.71%	85.71%	85.71%	91.73%
Full Fusion (AnemiaFusionNet)	87.88%	81.25%	92.86%	86.67%	93.98%

Key Observations

- Clinical dominance: standalone clinical attributes (RBC, MCV, PLT, MCHC — see the SHAP analysis in Section 8.3) provide a strong base predictive signal (89.85% AUC) on their own, consistent with their direct pathological relevance to anemia.
- Visual signal challenges: the image-only model collapses to majority-class prediction (100% recall but 42.42% precision/accuracy, 50.00% AUC), attributed to the limited dataset size (218 patients) causing the vision backbone to overfit.
- Geo-risk synergy: incorporating the geographic NFHS-5 prevalence prior via the ModalityFusionTransformer raises ROC-AUC from 91.73% (Image + Clinical) to 93.98% (Full Fusion) — the highest AUC among all four configurations — while recall also improves from 85.71% to 92.86%, at some cost to precision (85.71% → 81.25%).
- Full Fusion is the best-ranking configuration overall by AUC (93.98%), ahead of Clinical-only (89.85%) and Image + Clinical (91.73%), even though its point-classification accuracy (87.88%) is slightly below Clinicalonly's (90.91%) — the fusion model trades a small amount of raw accuracy for substantially better ranking quality and higher recall.

7. Performance Curves

7.1 ROC Curve

The plot below traces the true-positive rate against the false-positive rate for the Full Fusion model as its decision threshold is swept across all possible values. A curve that hugs the top-left corner indicates a classifier that can achieve a high true-positive rate while keeping false positives low; the diagonal would represent random guessing.

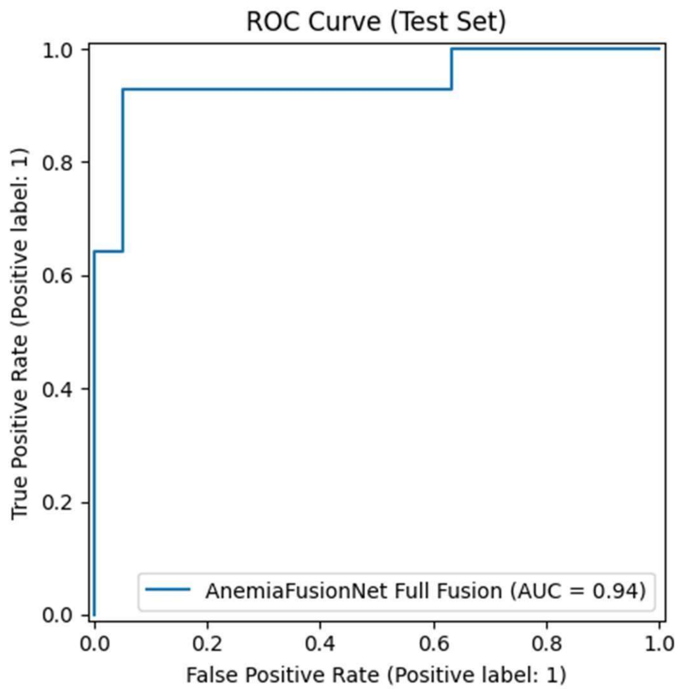

Figure 1. ROC curve for the Full Fusion AnemiaFusionNet model on the test set. The step shape reflects the small test set (n = 33); the plotted legend reports AUC = 0.94, matching the metrics export value of 93.98% for this configuration (Section 6).

7.2 Confusion Matrix

The heatmap below cross-tabulates actual patient labels (rows) against the model's predicted labels (columns) for the Full Fusion configuration; darker cells indicate a higher count of patients falling into that actual/predicted combination.

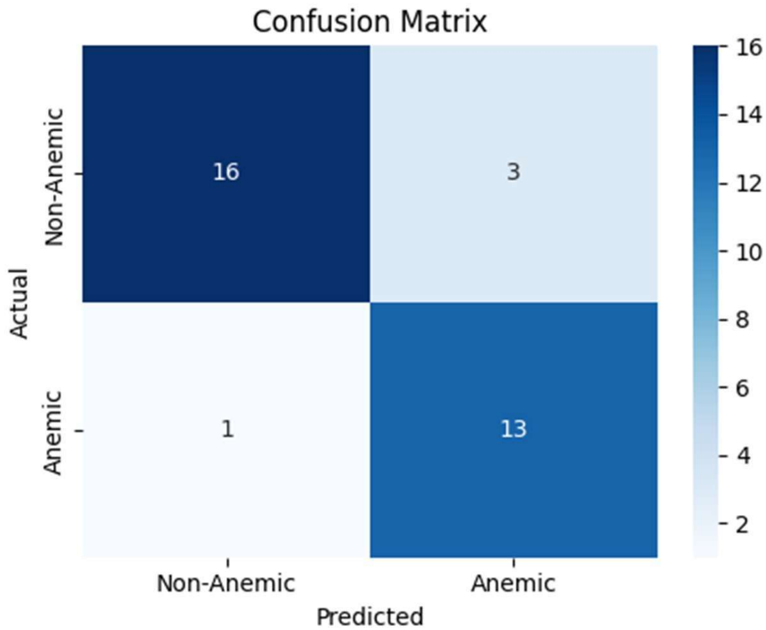

Figure 2. Test-set confusion matrix: 16 true negatives, 3 false positives, 1 false negative, 13 true positives (n = 33).

From the confusion matrix directly: accuracy = (16+13)/33 = 87.88%, precision = 13/(13+3) = 81.25%, recall = 13/(13+1) = 92.86%, and F1 = 86.67%. These figures match the Full Fusion row of the metrics export in Section 6 exactly, confirming the two artifacts describe the same evaluation run.

8. Model Explainability and Interpretability

8.1 Vision Branch — Grad-CAM

Grad-CAM was applied to the final convolutional layer of the EfficientNet-B0 vision branch to verify that the image classifier attends to clinically relevant anatomy rather than incidental image regions. The figure below shows two panels side by side: the original conjunctiva photograph on the left, and the same photograph overlaid with a GradCAM activation heatmap on the right, rendered with the standard JET color scale.

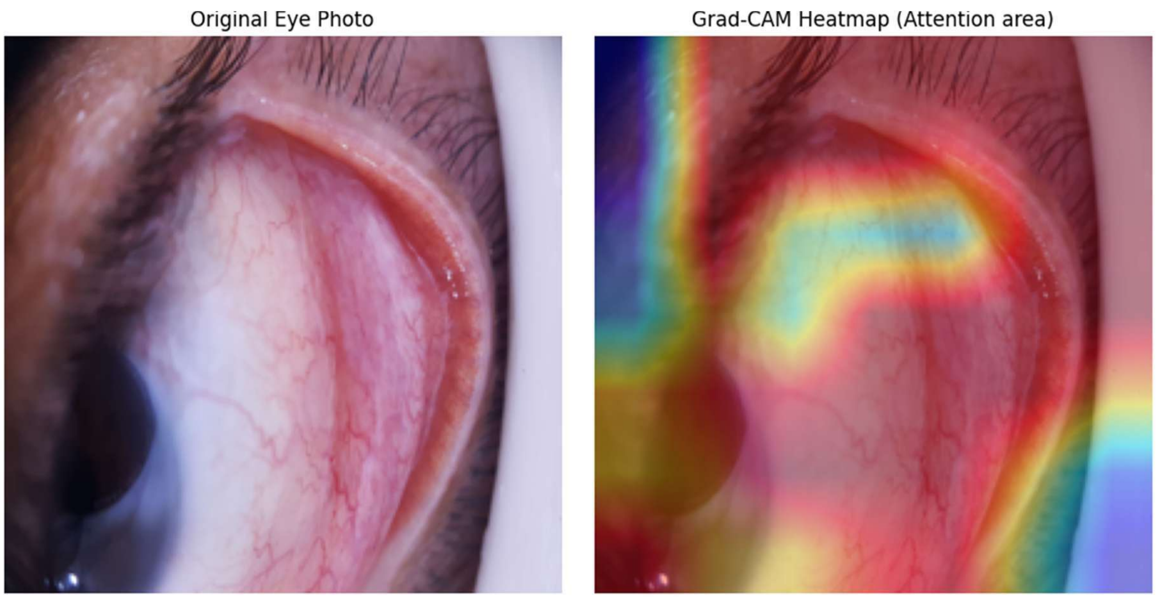

Figure 3. Left: original eye photo. Right: Grad-CAM heatmap overlay, JET color scale (red/orange/yellow = highest attention, green/light blue = transitional, dark blue/purple = no attention).

On the JET scale used here, red, orange, and yellow mark the regions the network weights most heavily in reaching its prediction; green and light blue mark regions of medium-to-low, transitional attention; and dark blue/purple mark regions the network effectively ignores. In this example, the warm red/orange band aligns closely with the palpebral conjunctiva — the inner lining of the lower eyelid where pallor is clinically assessed — while the surrounding skin, eyelashes, and pupil fall in the cool, near-zero-attention range. This indicates the vision backbone has learned to localize the correct anatomical structure for pallor assessment, even though, per Section 6, this branch's standalone predictive performance is currently poor due to limited training data.

8.2 Fusion Transformer — Modality Attention

Self-attention weights from the fusion transformer's [CLS] token were averaged across the test set to show how much each modality contributes to the pooled representation. The bar chart below plots one bar per modality (Image, Clinical, Geo-Risk) on the x-axis against its average attention weight on the y-axis.

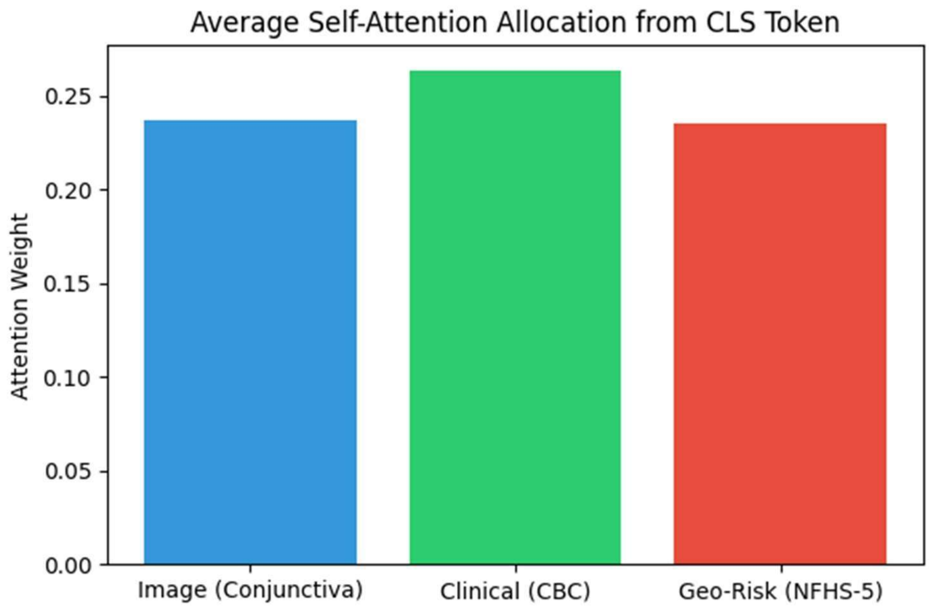

Figure 4. Average self-attention allocation from the [CLS] token: Clinical (CBC) ≈ 0.265, Image (Conjunctiva) ≈ 0.235, Geo-Risk (NFHS-5) ≈ 0.235 (bars do not sum to 1; remainder is attention retained on the [CLS] token itself).

The three modalities receive comparatively balanced attention, with clinical features drawing a modestly larger share (≈0.265) than image or geo-risk (≈0.235 each). This is broadly consistent with Section 6: the clinical branch is the strongest standalone predictor, but the transformer still draws meaningfully on the image and geo-risk branches — consistent with Full Fusion achieving the best overall ROC-AUC (93.98%) by combining all three signals rather than relying on clinical features alone.

8.3 Clinical Branch — SHAP Feature Importance

A SHAP KernelExplainer was applied to the standalone clinical MLP to attribute per-feature contributions to the anemia-positive class (Class 1). In the summary plot below, each row is one CBC feature (ordered top-to-bottom by overall importance), each dot is one test patient, the dot's horizontal position is that patient's SHAP value (positive = pushes toward an anemia prediction, negative = pushes toward healthy), and the dot's color encodes whether that patient's raw feature value was high (red/pink) or low (blue).

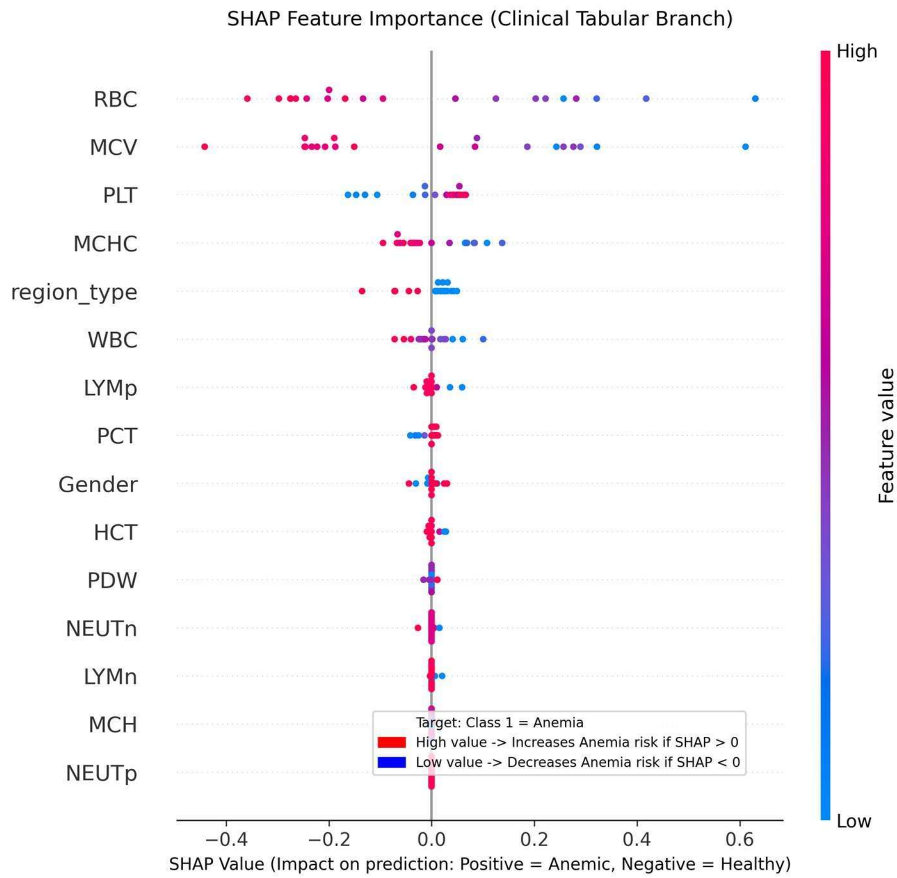

Figure 5. SHAP summary plot for the clinical branch, ranked from RBC (most influential, top) to NEUTp (least influential, bottom). Color encodes feature value (blue = low, red/pink = high).

RBC and MCV are the top-ranked features by SHAP magnitude, followed by PLT and MCHC. For RBC, low feature values (blue) generally push SHAP values positive (toward anemia risk), consistent with anemia's clinical definition as reduced red-cell mass. Below the top four, region_type, WBC, LYMp, and PCT show modest contributions, while Gender, HCT, PDW, NEUTn, LYMn, MCH, and NEUTp cluster tightly around zero SHAP value, indicating minor marginal contribution once RBC, MCV, PLT, and MCHC are accounted for.

9. Streamlit Web Application Interface

A Streamlit application (app.py) exposes the framework for interactive clinical demonstration, organized into three feature areas:

9.1 Real-Time Diagnostic Fusion Portal

- Upload of patient conjunctiva images (.jpg, .jpeg, .png) or selection of a sample record.
- Numeric inputs for CBC parameters (RBC, HCT, WBC, MCV, etc.) pre-filled with baseline medians.
- State dropdown showing NFHS-5 prevalence rates and normalized risk weights.
- On-click inference producing a binary classification (Anemic / Healthy) with confidence score.

9.2 Real-Time Explainability

- Live Grad-CAM heatmap overlay on the uploaded image.
- Self-attention rollout visualizing per-patient modality weighting dynamically.

9.3 Performance Dashboard

- Ablation tables, ROC curves, and confusion matrices rendered in-app.
- Interactive Mermaid rendering of the model architecture diagram.

10. Discussion, Data-Consistency Review, and Limitations

10.1 Metrics export and cross-artifact consistency

The figure below is the raw per-configuration metrics export underlying Section 6. It is corroborated by two independent checks: its Full Fusion row (accuracy 87.88%, precision 81.25%, recall 92.86%, F1 86.67%) can be reproduced exactly by hand from the confusion matrix in Figure 2, and its Full Fusion AUC of 93.98% closely matches the 0.94 shown on the plotted ROC curve's own legend in Figure 1. This cross-artifact agreement is the basis for treating the numbers in Section 6 as reliable for this evaluation run.

	accuracy	precision	recall	f1	auc
Image-only	0.4242424	0.4242424	1	0.5957446	0.5
Clinical-only	0.9090909	0.8666666	0.9285714	0.896552	0.8984962406015037
Image + Clinical	0.8787878	0.8571428	0.8571428	0.8571428	0.9172932330827067
Full Fusion (AnemiaFusionNet	0.8787878	0.8125	0.9285714	0.8666666	0.9398496240601505

Figure 6. Raw metrics export (spreadsheet screenshot): one row per configuration (Image-only, Clinical-only, Image + Clinical, Full Fusion) with columns for accuracy, precision, recall, F1, and AUC. This is the source used for Section 6.

10.2 Simulated tri-modal linkage

As detailed in Section 4, the clinical and geographic values attached to each patient are not their true, jointly measured values — they are label-conditioned or population-weighted samples. This means the fusion model's learned attention over the clinical and geo branches reflects how predictive those donor pools are of the coarse binary label in aggregate, not a validated patient-level multimodal relationship. Any claim of clinical deployability should be conditioned on eventually validating the architecture against a real tri-modal cohort.

10.3 Sample size and result stability

The image-derived cohort (218 patients; test set of 33 in the confusion matrix) is small for a deep vision backbone such as EfficientNet-B0, which plausibly explains the image-only branch's degenerate behavior. A test set of this size means that a handful of individual predictions can shift accuracy, precision, or AUC by several points — for example, the roughly 4-point AUC gap between Image + Clinical (91.73%) and Full Fusion (93.98%) reported in Section 6 corresponds to only a few patients' worth of ranking differences. Re-running the same architecture with a different random seed or train/test split could plausibly move these numbers by a comparable margin, so the relative ordering of configurations should be treated as suggestive rather than definitive without a larger test set or resampling-based confidence intervals.

11. Conclusion

AnemiaFusionNet demonstrates a technically coherent tri-modal fusion architecture — independent vision, tabular, and geo-risk encoders combined through a self-attention transformer — with explainability instrumentation (GradCAM, attention rollout, SHAP) covering every branch of the model. On the current evaluation run (Section 6, corroborated in Section 10), clinical CBC features provide the strongest single-modality signal, the image-only branch collapses to majority-class prediction on the current cohort size, and the full tri-modal fusion achieves the best overall ranking performance (93.98% ROC-AUC) — ahead of Image + Clinical (91.73%) and Clinical-only (89.85%) — while trading a small amount of raw accuracy for improved recall. Given the small test set (n = 33), this ranking should be treated as a promising signal rather than a settled result.